# [Working Title] Your English

*formerly Colour Your English (CYE)*

## Project Bible

*A living document for design decisions, scientific grounding, and open questions*

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# How to Use This Document

This is a living project bible — not a finished specification, not a pitch deck, not an academic paper. It captures where the project is NOW: what we know, what we've decided, what's grounded in research, and what's still open.

Any instance of Claude working with Sarah on CYE should read this first. It won't tell you everything. It will tell you enough to not start from scratch.

**For the next Claude instance:** Read Sections 1–3 before doing anything. Read the Open Questions before proposing anything. Read the Sources before citing anything.

# 1. What CYE Is

## 1.1 The Core Idea

CYE is a pedagogical system that teaches English grammar through synaesthetic principles — colour, texture, sound, and spatial relationships. Students don't learn rules and then apply them. They experience grammatical categories as sensory properties, building intuitive pattern recognition before (or alongside) explicit knowledge.

This isn't decoration. It's grounded in Cognitive Load Theory: multi-sensory input opens parallel processing channels, reducing the load on any single channel. The colour IS the grammar. The texture IS the function. There is no legend to cross-reference, no key to look up.

## 1.2 The Three Rings

**Ring 1 — Core:** A teaching tool. Interactive activities using colour, texture, and sound to teach specific grammar points (phrasal verbs, tenses, adjective order, etc.) to L2 English learners. This is the prototype. This is what gets built first.

**Ring 2 — Framework:** A design methodology for multi-sensory learning. The Sensory Vault, the Climate Zones, the Warp/Weft metaphor. Applicable beyond English, beyond grammar, beyond language teaching. This is the system.

**Ring 3 — Horizon:** A civilisational hypothesis. If language shapes perception (Sapir-Whorf), then raising children with dual-mode symbolic language — sequential AND simultaneous, spoken AND patterned — could fundamentally change communication?

*Working principle: Build Ring 1 first. Let Ring 2 emerge from the building. Let Ring 3 remain a guiding star, not a distraction.*

### 1.3 The Name Problem

"Colour Your English" is taken (Hirschman & Bautz, Perceptia Press). A new name is needed. The name should convey sensing language, not just colouring it. Candidates explored and shelved: "Pigment Your English" (too close), "English Loom" (beautiful but loses the sensory core), "EnglishScape" (no).

## 2. Design Decisions Made

### 2.1 The Sensory Vault

The Sensory Vault is NOT a universal colour-grammar dictionary. It is a set of design principles for assigning sensory mappings within any given activity.

**Constraint:** Lindsey & Brown (2014) established a perceptual ceiling of ~15–16 reliable colour categories in English speakers. CYE activities use 3–4 colour mappings maximum per activity. Exceeding this hits the perceptual ceiling and creates confusion, not learning.

**Principle:** Warm colours for semantic content (nouns, verbs, adjectives — things that carry meaning). Cool colours for structural/functional elements (prepositions, conjunctions, articles — things that organise meaning). Textures for weight/temporality (rough/heavy for past, smooth/light for future). Sound for movement/direction (rising for questions, falling for statements).

**Key insight:** Colour mappings are activity-specific, not universal. The grammar mindmap (Basic English Grammar) has hundreds of potential categories. Each activity picks a small meal from the menu. The mindmap is the menu, not the plate.

### 2.2 Climate Zones (Expertise Progression)

CYE uses tiered zones that scaffold differently based on learner expertise. This is grounded in Kalyuga's Expertise Reversal Effect: instructional designs that help novices can actively hinder advanced learners.

**Lollipop Playground:** Heavy colour-coding, detailed guidance, full sensory scaffolding. For A1–A2 learners. Provides external executive function that novices lack internally.

**The Meadow:** Reduced scaffolding. Student becomes "Gardener" — chooses how to "plant" knowledge. The Third Space where fixed grammar rules (Warp) meet student choice (Weft). For B1–B2 learners.

**The Architecture:** Minimal colour-coding. Scaffolds faded. Student has built internal schemas; redundant support wastes cognitive resources. For C1+ learners.

**When to fade:** When external guidance begins to overlap with student's internal knowledge, forcing them to cross-reference both. Use rapid diagnostic methods to detect this transition in real-time.

## 2.3 The Third Space

In pedagogy, the Third Space is where the student's home literacy (Space 1) meets the school's formal structures (Space 2). In CYE:

**The Warp (Space 2):** Fixed grammar rules. The "must-have" structure. Non-negotiable.

**The Weft (Space 1):** Student's choice. Their vocabulary, their hue, their personal connection to the language.

The student weaves Warp and Weft together, creating something that is both structurally correct and personally meaningful. This moves them from "data entry clerk" to active creator negotiating their own learning identity.

## 2.4 Dyslexia and Accessibility

CYE's design inherently benefits dyslexic learners, but this must be intentional, not accidental:

**Visual attention span:** Dyslexic learners struggle with text-heavy visual search. CYE transforms text-search tasks into pattern-recognition tasks, reducing anxiety and fatigue.

**3D sculptures:** Rotating a phrasal verb to see the particle from different angles reduces "clutter" that overloads visual attention.

**Font and colour choices:** Consistent use of dyslexia-friendly fonts, carefully chosen colour contrasts, avoidance of pure white backgrounds. Sarah's existing practice uses teal-orange schemes throughout.

*Source: Multilingualism, Literacy and Dyslexia — Sarah is currently reading this. Chapters on visual processing and multisensory approaches are most relevant.*

## 2.5 Teal and Orange

The consistent teal-orange colour scheme across all Sarah's materials isn't arbitrary. Teal is one of four emerging colour terms in American English (Lindsey & Brown, 2014). It sits at the green-blue boundary — a liminal, transitional colour. Orange is warm, active, alive. Together they represent the two poles of CYE: cool structure and warm content, architecture and flight, the room and the detonation that happens in it.

# 3. Scientific Grounding

## 3.1 Source Map

| Source | CYE Relevance | Key Pages | Status |
|---|---|---|---|
| Sweller — Managing Cognitive Load in Adaptive Multimedia Learning (2008) | Split-attention effect: colour and category cannot be separate things to mentally reunite. Modality effect: audio + visual + texture opens parallel processing channels. | p.43 (split-attention), p.46 (modality effect), pp.34–53 (theory) | Read ✓ |
| Lindsey & Brown — The Color Lexicon of American English (2014) | Perceptual ceiling of ~15–16 colour categories. Teal as emerging term. Gender differences in colour vocabulary. Constrains CYE palette design. | Full paper read | Read ✓ |
| Multilingualism, Literacy and Dyslexia (2nd ed.) | Visual attention span, multisensory literacy approaches, Third Space concept. | p.9 (visual attention), p.13 (anxiety/fatigue), p.127 (Third Space) | Reading |
| Kalyuga — Expertise Reversal Effect | Validates Climate Zones: novice scaffolding hinders experts. Fading guidance. Rapid diagnostics. | pp.58–65 (expertise reversal), pp.68–81 (fading/diagnostics) | Via NotebookLM |
| Sapir-Whorf Hypothesis (various) | Theoretical backbone: language shapes perception. If colour cognition is shaped by language, building new colour-language creates new perception. | General | Ongoing |
| Chiang — Story of Your Life (1998) | Fictional backbone: semasiographic writing system restructures temporal perception. Dual-mode cognition. The heptapod sentence. | Full story read | Read ✓ |
| Basic English Grammar mindmap | Comprehensive taxonomy of grammar categories. The “menu” from which each activity picks 3–4 items. Shows scope of potential content. | Full map reviewed | Reviewed ✓ |
| Sarah’s “Ocean Meeting the Human Shore” | Early Bridge/Third Space thinking. Compression problem, shared latent space, synesthetic synthesis, grounding crisis. Proto-CYE theory. | Full document | Reviewed ✓ |

## 3.2 Key Scientific Hooks

### Split-Attention Effect (Sweller, p.43)

When text and its referent are spatially separated, learners waste cognitive resources mentally integrating them. CYE’s solution: the colour IS the category. No legend, no key, no cross-referencing. The mapping is the learning. This is the single most important design constraint.

### Modality Effect (Sweller, p.46)

Information distributed across auditory and visual channels is processed more effectively than information in a single channel. This is the scientific licence for the Sensory Vault: audio textures

accompanying visual colour isn't decoration, it's opening parallel processing pipes. Working memory capacity expands when channels are distributed.

### Perceptual Ceiling (Lindsey & Brown, 2014)

English speakers reliably use ~15–16 colour categories. After this, consensus collapses. CYE cannot have 30 colours. The constraint: 3–4 per activity, activity-specific mappings, no universal colour-grammar dictionary. This is a hard engineering limit, not a preference.

### Expertise Reversal Effect (Kalyuga)

The same scaffolding that helps a novice actively hinders an expert. Heavy colour-coding is essential at A1 and counterproductive at C1. The Climate Zones aren't just a nice progression — they're a cognitive necessity. Scaffolding must fade as internal schemas develop.

### Third Space (Peer & Reid)

The pedagogical Third Space is where home literacy meets school structure. In CYE, this becomes the Warp/Weft metaphor: fixed grammar rules woven with student choice. The student as Weaver/Gardener, not data entry clerk.

### Visual Attention Span (Peer & Reid, p.9)

Critical for dyslexic learners. Text-heavy environments cause anxiety and fatigue. CYE's 3D sculptures and colour-coding transform text-search into pattern-recognition, directly addressing this.

# 4. Theoretical Framing

## 4.1 The Sapir-Whorf Connection

CYE's deepest hypothesis: if language shapes perception (and research shows it does, measurably, for colour cognition), then building a new sensory language for grammar creates new perceptual capabilities. Students don't just learn grammar differently — they perceive language differently.

## 4.2 The Chiang Frame

Ted Chiang's "Story of Your Life" provides the fictional backbone. Louise Banks learns Heptapod B — a semasiographic writing system where meaning is non-sequential and simultaneous. Learning the notation restructures her temporal perception. She experiences memory, present, and future simultaneously.

CYE operates on a parallel principle at a smaller scale: learning to perceive grammar through colour/texture/sound doesn't just add a mnemonic layer. It creates a parallel processing mode. The student who has internalised colour-grammar mappings doesn't translate FROM colour TO grammar; they perceive both simultaneously. Dual-mode cognition.

# 5. Open Questions

| Open Question | Stream | Priority |
|---|---|---|
| What should the project be called? (CYE name is taken) | Naming | High |
| Should colour mappings use any universal principles across activities, or be entirely activity-specific? | Sensory Vault | High |
| How to detect the fade point between Climate Zones in real-time? What rapid diagnostic methods work in a classroom of 25? | Climate Zones | High |
| What does the Core prototype look like? Which grammar point first? Phrasal verbs? Tenses? Adjective order? | Ring 1 | High |
| How to handle students with colour vision deficiency? Texture/sound as primary channels? | Accessibility | Medium |
| What platform? Standalone app, Moodle plugin, web-based, or hybrid? | Technical | Medium |
| How does the Sensory Vault interact with L1 colour associations? (French colour terms vs English colour terms) | Cross-lingual | Medium |
| What team/collaborators does Sarah need? Developers, cognitive scientists, dyslexia specialists? | Practical | High |
| How to document this for academic publication? Which journal? What framing? | Output | Medium |

# 6. Working Architecture

## 6.1 Work Streams

**Stream 1 — Teaching Materials:** Vocab quizzes, HTML games, H5P activities, Moodle content. Self-contained tasks. Low continuity cost. Any Claude instance can handle these with minimal context. Use existing skills (vocab-quiz-builder, html-game-builder).

**Stream 2 — CYE Development:** This document. The spec. Prototype design. Scientific grounding. Naming. Needs continuity but can survive instance changes IF this bible is kept current.

## 6.2 Key Files

**Basic_English_Grammar.pdf:** Grammar taxonomy mindmap. The “menu.”

**Lindsey & Brown (2014):** Colour perception constraints paper.

# 7. Next Steps

These are the immediate priorities, in order:

**1.** Choose a prototype grammar point for the Core activity (Ring 1). Recommendation: phrasal verbs or present perfect tense — both are high-pain points for L2 learners and map well to sensory encoding.

**2.** Design the Sensory Vault principles for the chosen grammar point: which 3–4 colour/texture/sound mappings, and why.

**3.** Build a working prototype (HTML or React). Test with students. Iterate.

**4.** Continue reading the dyslexia source. Extract specific design constraints for accessibility.

**5.** Let the name arrive. Don't force it.